

Agilex 5 Camera 4Kp60 HDR — Lab Bring-up Guide

Field	Value
Design	4Kp60 Multi-Sensor HDR Camera Solution System Example Design
Release	Quartus Prime Pro 26.1 / repo branch <code>rel/26.1</code>
Hardware	Agilex 5 FPGA E-Series 065B Modular Development Kit
Cameras	Framos FSM:GO IMX678 (1 or 2 modules)
Audience	Lab / engineering team
Doc status	Lab working procedure + notes from bring-up

1. Purpose

This document tells a teammate how to **boot and run** the pre-built Altera camera design on the Modular Development Kit, open the web GUI, and get camera video on DisplayPort.

It has two layers:

1. **Official Altera flow** (from the design documentation)
 2. **Lab-specific notes** discovered during bring-up on this host PC (especially Ethernet MTU)
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2. Customer demo fast start (pre-provisioned kit)

Use this section when the kit has already been prepared and you need to bring it up in front of a customer. It is deliberately short; use the detailed procedure in Section 7 if a step fails.

What must already be ready

Item	Required state
QSPI	<code>top.core.jic</code> was programmed once successfully
microSD card	Contains the matching <code>hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic</code> image
Boot switches	SOM SW1 is ON-ON (QSPI / AS×4 boot)
Camera	Framos IMX678 connected to MIPI0 or MIPI1, with pin 1 aligned
Display	Carrier DisplayPort TX connected to the selected monitor

Item	Required state
Host connections	Board J6 Ethernet and SOM J2 USB connected to the host PC
Host network	Board and host are on the same Ethernet network; VPN is disabled if it interferes

A prepared microSD card alone is not enough. QSPI must already contain the matching `top.core.jic` boot image.

Demo sequence

1. Power on

Verify the connections above, insert the microSD card, and power on the board.

2. Wait for application startup

Wait about two minutes for Linux and the camera application to start.

3. Open the HPS serial console

Use 115200 8N1, press Enter, and log in as `root` :

```
sudo minicom -D /dev/ttyUSB6 -b 115200
```

`ttyUSB6` is the usual lab mapping; if it is blank, use the serial-port guidance in Section 7.3.

4. Get the board address

```
ip -4 addr show eth0
```

5. Open the GUI on the host PC

```
http://<board-ip>/
```

6. Select the camera input

In the web GUI, select **Input Config** → **Input Source** → **Camera**.

7. Confirm video

Confirm camera video appears on the DisplayPort monitor.

If the GUI does not open

First test normal Ethernet operation:

```
ping -c 2 -M do -s 1472 <board-ip>
```

- If this succeeds, keep the default MTU 1500 and continue with the GUI.
- If this fails, apply the **lab-only MTU 400 workaround** in Section 8, then reload the GUI.

If the monitor has no video

In the GUI select **Input TPG** first. If the test pattern appears, switch back to **Camera**. For the MIPI1 single-camera known issue, toggle **Camera** → **Input TPG** → **Camera**.

3. Official references

Resource	Link
Design overview	https://altera-fpga.github.io/rel-26.1/embedded-designs/agilex-5/e-series/modular/camera/camera_4k/camera_4k/
Detailed guide	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/docs/es/camera/camera_4k/camera_4k.md
Repository	https://github.com/altera-fpga/agilex5-ed-camera/tree/rel/26.1
Pre-built binaries	https://github.com/altera-fpga/agilex5-ed-camera/releases/tag/rel-26.1-isp_hdr-MDK_RevB_GrpB
Repo quick start	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/AGX_5E_Altera_Modular_Dk_ISP_designs/HDR_CAMERA.md
Known issues	https://github.com/altera-fpga/agilex5-ed-camera/blob/rel/26.1/docs/es/camera/camera_4k/known_issues.md

4. Does the official design require setting MTU?

No. Altera's official camera 4K documentation does **not** ask you to change Ethernet MTU.

Official network guidance is only:

- Connect board **J6** Ethernet to the same network as the host PC (direct cable, switch, or router)
- Disable VPN on the host if it interferes
- Get the board IP with `ifconfig / ip a` after login as `root`
- Open `http://<board-ip>/` in a browser

What if you do not set MTU?

On a **healthy** Ethernet path (normal 1500-byte frames work):

- You do **nothing** with MTU
- GUI and board work at default MTU 1500

On **this lab host path** (observed during bring-up):

- Frames larger than ~400 bytes were dropped (direct cable, switch, and USB-Ethernet dongles all showed the same ceiling)
- Symptoms if you leave MTU at 1500:

- `ping` small packets OK
- `ping -M do -s 1472` fails
- Browser page hangs / blank
- `curl` may show HTTP headers (200 OK , Content-Length: 1112) but **0 bytes body** / timeout
- SSH key exchange can stall
- Workaround used in this lab: set **MTU 400 on both PC and board**
- Side effect of MTU 400: web GUI may briefly grey out every few seconds (WebSocket/UI traffic is fragmented). Live video is on **DisplayPort**, not the browser.

Team rule: First test large ping. Only apply MTU 400 if large frames fail.

```
# Replace BOARD_IP with the board address
ping -c 2 -M do -s 1472 BOARD_IP
```

Result	Action
Large ping succeeds	Keep default MTU 1500 (official / preferred)
Large ping fails	Use lab MTU 400 workaround (Section 8)

5. One-time programming (already completed on this kit)

Repeat only if microSD or QSPI is wiped.

Required release files

From release `rel-26.1-isp_hdr-MDK_RevB_GrpB` :

File	Purpose
<code>top.core.jic</code>	QSPI boot image
<code>hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz</code>	microSD Linux + app image

Burn microSD

```
gunzip -k hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic.gz
# Identify the whole-disk device carefully (e.g. /dev/sdX), not a partition
sudo dd if=hps-first-vvp-isp-demo-image-agilex5_mk_a5e065bb32aes1.wic \
  of=/dev/sdX bs=1M status=progress conv=fsync
sync
sudo eject /dev/sdX
```

Insert the card in the **SOM** microSD slot.

Program QSPI once

1. Power **off**

2. SOM **SW1 = OFF-OFF** (MSEL = JTAG)
3. Power **on**
4. Program:

```
jtagconfig
quartus_pgm -c 1 -m jtag -o "pvi;top.core.jic"
```

1. Power **off**
2. SOM **SW1 = ON-ON** (MSEL = ASx4 / QSPI boot)

6. Hardware checklist (every session)

Item	Setting / connection
Boot mode	SOM SW1 = ON-ON
microSD	Inserted in SOM slot
Ethernet	Board J6 → host PC Ethernet (lab preferred: direct shared link)
USB	SOM J2 + Carrier J35 both to PC
Cameras	Framos IMX678 on MIPI0 and/or MIPI1, pin 1 aligned
Display	Carrier DisplayPort TX → 4K-capable monitor (prefer native DP cable)

USB serial devices (`/dev/ttyUSB*`) appear only when the board is powered on.

7. Every-boot procedure (lab recommended)

7.1 Boot the host PC first

Configure Wired Ethernet as **Shared to other computers**.

```
ip -4 addr show enp3s0
# Expected on this lab PC: inet 10.42.0.1/24
```

If sharing is not active (connection name on this PC is `netplan-enp3s0`):

```
sudo nmcli connection modify "netplan-enp3s0" ipv4.method shared
sudo nmcli connection up "netplan-enp3s0"
ip -4 addr show enp3s0
```

7.2 Power on the board

Wait about **2 minutes** for Linux and `VvpIspDemo` to start.

7.3 Open HPS serial console

With **J2 + J35** connected you typically get **8** USB serial ports.

```
ls -l /dev/ttyUSB*

gnome-terminal --title="HPS" -- bash -c \
  "sudo minicom -D /dev/ttyUSB6 -b 115200; exec bash"
```

- Settings: **115200 8N1**, hardware flow control **OFF**
- Press **Enter**
- Login: `root` (no password)

Serial port map (important)

Port group (typical)	Role
<code>ttyUSB0 - ttyUSB3</code>	JTAG / Nios group
<code>ttyUSB2</code>	Often Nios ("WELCOME TO INTEL NIOS SYSTEM") — not Linux
<code>ttyUSB4 - ttyUSB7</code>	HPS UART group
<code>ttyUSB6</code>	Often HPS Linux (3rd port of HPS group)

`ttyUSB` numbers can change after reboot. If `ttyUSB6` is blank, try `ttyUSB4 / 5 / 7`, or open all four and power-cycle once with minicom already open.

7.4 Read board IP

```
ip -4 addr show eth0
# Typical on shared link: 10.42.0.212 (can change)
```

If no IPv4 address:

```
udhcpc -i eth0
ip -4 addr show eth0
```

Always re-read the IP. Do not assume it stays `.212`.

7.5 Network health check (decide MTU)

On the host:

```
BOARD_IP=10.42.0.212 # replace with actual eth0 IP

ping -c 2 $BOARD_IP
ping -c 2 -M do -s 1472 $BOARD_IP
```

- If **1472 succeeds** → skip Section 8, keep MTU 1500
- If **1472 fails** → apply Section 8 (MTU 400)

7.6 Open the web GUI

```
curl -sS --max-time 10 http://$BOARD_IP/ | wc -c
# Healthy response size is about 1112 bytes of HTML

google-chrome-stable http://$BOARD_IP/
```

You should see the Altera splash, then the camera GUI.

7.7 Select camera input

In the GUI:

1. Open **Input Config**
2. Set **Input Source** → **Camera**
3. If video is missing (especially single camera on MIPI1): toggle
Camera → **Input TPG** → **Camera**
(official known-issue workaround)

Optional board log check:

```
grep -iE 'Found camera|Active input|GMSL' /tmp/vvp-isp-demo.log | head -n 30
```

Healthy example:

- Found camera: FSM-IMX678 as Cam 0 / Cam 1
- Active input: Camera 0
- 3840x2160p @60

8. Lab-only MTU workaround

Use only when large ping fails on your host↔board path.

Board

```
ip link set eth0 mtu 400
```

Host PC (direct shared link on `enp3s0`)

```
sudo ip link set enp3s0 mtu 400
sudo ip route replace 10.42.0.0/24 dev enp3s0 src 10.42.0.1 mtu lock 400
```

Verify

```
ping -c 2 $BOARD_IP
curl -sS --max-time 10 http://$BOARD_IP/ | wc -c
# expect ~1112
```

Optional size probe

```
for s in 56 200 400 500 800 1200 1472; do
  if ping -c 1 -W 1 -M do -s $s $BOARD_IP >/dev/null 2>&1; then
    echo "OK $s"
  else
    echo "FAIL $s"
  fi
done
```

On the lab path that needed this workaround, sizes above ~400 failed.

9. DisplayPort blank / “please load image”

Cameras can be detected in software while the monitor stays blank. That is usually a **DisplayPort link** issue, not a sensor detect failure.

1. In GUI, set **Input Source = Input TPG** - If TPG appears → DP works; switch back to **Camera** - If still blank → check cable / TX connector / monitor input
 2. Prefer native **DP→DP**; cheap DP→HDMI adapters often fail at 4Kp60
 3. Use carrier **DisplayPort TX**
 4. Reseat cable; power-cycle monitor
 5. Optional: watch Nios console (`tttyUSB2`) while replugging DP for link messages
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10. Useful board commands

```
ps | grep VvpIspDemo
tail -n 50 /tmp/vvp-isp-demo.log

# Confirm embedded web server responds locally
wget -S -O /tmp/local.html -T 5 http://127.0.0.1/
wc -c /tmp/local.html

# Restart application if needed
killall VvpIspDemo
cd /home/root && ./start.sh
```


App location: `/home/root/VvpIspDemo` (started by `/home/root/start.sh`).

11. What not to redo every boot

- Do **not** rewrite the microSD image
 - Do **not** reprogram QSPI (`top.core.jic`) unless flash was erased or boot is broken
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12. Troubleshooting quick table

Symptom	Likely cause	Action
No <code>/dev/ttyUSB*</code>	Board off / USB unplugged	Power on; connect J2 and J35
Only 4 ttyUSB ports	One USB cable missing	Connect both J2 and J35
Only Nios banner	Wrong serial port	Use HPS group's 3rd port (often <code>ttyUSB6</code>)
Ping OK, browser blank / curl body = 0	MTU / large-frame path issue	Run 1472 ping test; apply Section 8 if needed
IP not <code>.212</code>	DHCP lease changed	Re-read <code>ip -4 addr show eth0</code>
Cameras found, monitor blank	DP link	Try Input TPG; check DP cable/TX
GUI greys out every few seconds	MTU 400 side effect	Accept for control, or fix Ethernet path for MTU 1500
VPN connected	Host network interference	Disable VPN (official guidance)

13. One-page cheat sheet

1. PC Wired = Shared -> `10.42.0.1`
 2. Power board -> SW1 ON-ON
 3. minicom `/dev/ttyUSB6 115200` -> root
 4. Board: `ip -4 addr show eth0`
 5. Host: `ping -M do -s 1472 BOARD_IP`
 - fail -> set MTU 400 on board + PC
 - pass -> keep MTU 1500
 6. Chrome `http://BOARD_IP/`
 7. GUI Input Config -> Camera
(if needed: Camera <-> TPG <-> Camera)
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14. Document history

Version	Notes
1.0	Initial lab bring-up capture
1.1	Reformatted for team use; clarified that MTU is not an official Altera requirement and is a lab-path workaround only
1.2	Added customer demo fast-start section for a pre-provisioned board and microSD card